

KUMEJIMA, Japan

WMO: 479290

Lat: 26.3381N

Lon: 126.8039E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	11.9	2.3	4.5	12.7	3.5	4.9	13.5	9.5	16.5	8.6	16.5	3.4	330	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	32.5	27.4	32.0	27.2	31.4	27.0	28.2	31.2	27.8	30.8	27.5	30.5	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.2	30.2	27.0	22.7	30.0	26.6	22.2	29.7	90.7	31.3	89.1	31.0	87.5	30.6	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.2	7.2	DB	8.1	33.2	1.2	0.7	7.2	33.7	6.5	34.1	5.8	34.5	4.9	35.0	
			WB	6.0	28.5	1.2	0.5	5.1	28.9	4.3	29.2	3.7	29.5	2.8	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	17.0	17.4	18.9	21.6	24.2	27.3	29.2	27.9	25.5	22.5	18.9		
		DBStd	4.84	2.63	2.85	2.83	2.36	1.98	1.96	1.04	1.08	1.44	1.74	2.51		
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	162	59	47	28	3	0	0	0	0	0	1	23		
		CDD10.0	4863	216	208	277	348	441	518	596	536	481	376	277		
		CDD18.3	1983	16	22	47	101	183	268	338	332	286	223	127		
		CDH23.3	17853	6	14	58	290	1063	2765	4247	4124	3175	1679	396		
(14)	Wind	CDH26.7	5807	0	0	1	14	131	837	1814	1711	1026	254	19		
		WSAvg	3.9	4.0	4.0	4.0	3.9	3.4	3.7	3.8	3.8	3.7	4.1	4.0		
		PrecAvg	2179	140	140	184	196	260	278	135	197	225	153	126		
		PrecMax	3412	402	441	430	523	590	645	412	615	1263	587	360		
		PrecMin	1460	39	28	45	69	31	77	0	31	32	14	34		
		PrecStd	439	71	87	90	101	130	130	122	149	236	132	74		
		DB	24.2	25.0	26.2	28.2	30.3	32.2	33.1	33.3	32.3	30.8	28.4	25.6		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	20.5	21.4	22.6	24.7	26.5	27.8	27.6	27.5	27.3	26.3	24.6	22.0		
		DB	22.9	23.5	24.8	26.9	29.1	31.4	32.5	32.5	31.5	29.7	27.2	24.3		
		MCWB	19.5	20.4	21.5	23.9	25.7	27.5	27.4	27.3	26.9	25.6	23.4	20.5		
		DB	21.8	22.6	23.8	25.9	28.1	30.8	32.0	31.9	30.9	28.9	26.2	23.4		
		MCWB	18.4	19.7	20.8	23.2	25.0	27.2	27.2	27.1	26.5	24.9	22.7	19.8		
		DB	20.9	21.6	22.9	25.0	27.2	30.1	31.5	31.3	30.3	28.1	25.4	22.6		
		MCWB	17.5	18.7	20.0	22.4	24.4	26.9	27.0	26.9	26.1	24.2	22.0	19.0		
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	21.6	22.2	23.3	25.3	27.0	28.5	28.7	28.5	28.2	27.1	25.4	23.1		
		MCDB	23.5	24.1	25.3	27.2	29.3	31.2	31.7	31.5	31.1	29.7	27.6	24.9		
		WB	20.2	21.3	22.4	24.4	26.2	27.9	28.2	28.1	27.6	26.4	24.3	21.6		
		MCDB	22.1	23.1	24.3	26.3	28.3	30.5	31.3	31.2	30.5	28.9	26.4	23.5		
		WB	19.1	20.3	21.5	23.7	25.6	27.5	27.8	27.7	27.1	25.7	23.5	20.6		
		MCDB	21.4	22.1	23.4	25.5	27.6	30.0	30.9	30.8	29.9	28.2	25.7	22.9		
		WB	18.0	19.2	20.6	23.0	25.0	27.1	27.4	27.4	26.7	25.1	22.7	19.5		
(34)	Mean Daily Temperature Range	MCDB	20.4	21.2	22.5	24.7	26.8	29.6	30.5	30.4	29.4	27.4	25.0	22.1		
		MDBR	4.3	4.5	4.9	4.7	4.5	4.1	4.5	4.6	4.5	4.1	3.9	4.1		
		5% DB	5.5	5.7	5.7	5.2	4.9	4.3	4.9	5.1	4.9	4.7	4.6	5.0		
		MCWBR	4.1	4.1	4.2	3.5	2.9	2.2	2.2	2.2	2.4	2.5	3.1	3.6		
		5% WB	4.8	5.0	5.2	4.8	4.5	4.2	4.6	4.5	4.6	4.2	4.3	4.5		
		MCWBR	4.4	4.4	4.7	3.8	3.0	2.3	2.3	2.2	2.3	2.4	3.2	3.9		
		MCWBR	4.4	4.4	4.7	3.8	3.0	2.3	2.3	2.2	2.3	2.4	3.2	3.9		
(40)	Clear-Sky Solar Irradiance	taub	0.471	0.501	0.575	0.582	0.523	0.482	0.475	0.491	0.488	0.487	0.475	0.468		
		taud	2.091	2.032	1.868	1.912	2.135	2.311	2.329	2.275	2.252	2.189	2.174	2.104		
		Ebn at Noon	761	772	738	746	789	816	821	809	802	777	754	743		
		Edh at Noon	142	162	201	197	157	131	129	135	135	137	130	135		
(44)	All-Sky Solar Radiation	RadAvg	2.22	2.80	3.55	4.24	4.80	5.34	6.56	5.99	5.17	3.99	2.87	2.30		
		RadStd	0.30	0.47	0.39	0.46	0.66	0.49	0.55	0.42	0.45	0.36	0.27	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.27	N/A	N/A	+0.39	+0.54	+0.60	N/A	N/A	+105	+73
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	+0.44	+0.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page